

Programming Assignment Report

Design and Analysis of Algorithms

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Problem Name: A very big sum

Problem Statement: In this challenge, you need to calculate and print the sum of elements in an array, considering that some integers may be very large.

Function Description

Complete the function A Very Big Sum with the following parameter(s):

- Int ar[n]: an array of integers

Return

- long: the sum of the array elements
- **Input Format**
- The first line of the input consists of an integer .

The next line contains space-separated integers contained in the array.

Output Format

- Return the integer sum of the elements in the array.

Problem HackerRank Link:

<https://www.hackerrank.com/challenges/a-very-big-sum/problem?isFullScreen=true>

Github Submission Link:

<https://github.com/rohanagrawal2706-blip/DAA-LAB/blob/main/A%20very%20big%20sum>

Solution Steps: Algorithm-Big Sum

1. **Start**

2. **Create** a scanner object to take input from the user.
3. **Read** an integer N (number of elements).
4. **Initialize** a variable sum of type long and set it to 0.
5. **Repeat while** N > 0:
 - Read the next integer from input.
 - Add the integer to sum.
 - Decrease N by 1.
6. **Print** the value of sum.
7. **Stop**

Complete JAVA Code:

```
import java.util.Scanner;
```

```
public class BigSum {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
  
        int N = scanner.nextInt();  
        long sum = 0;  
        while (N-- > 0 ) {  
            sum += scanner.nextInt();  
        }  
        System.out.println(sum);  
    }  
}
```

Observation/Solution:

- 1. The program imports the Scanner class to allow input from the user.**
- 2. A Scanner object is created to read values from the keyboard.**
- 3. The program reads an integer N, which represents the number of values to be entered.**
- 4. A variable sum of type long is initialized to 0 to store the total of the numbers.**
- 5. A while loop is used with the condition $N > 0$ to repeat the process N times.**
- 6. Inside the loop, the program reads the next integer from the user.**
- 7. The entered number is added to the variable sum.**
- 8. The value of N is decreased by 1 after each iteration ($N--$).**
- 9. The loop continues until all N numbers are processed.**
- 10. Finally, the program prints the total sum of all the entered numbers using `System.out.println(sum)`.**